



A lightweight hybrid CNN-LSTM explainable model for ECG-based arrhythmia detection

Negin Alamatsaz^a, Leyla Tabatabaei^a, Mohammadreza Yazdchi^{a,*}, Hamidreza Payan^b,
Nima Alamatsaz^{c,d}, Fahimeh Nasimi^{a,e}

^a Department of Biomedical Engineering, Faculty of Engineering, University of Isfahan, Isfahan, Iran

^b Department of Electrical Engineering, Amirkabir University of Technology, Tehran, Iran

^c Department of Biomedical Engineering, New Jersey Institute of Technology, Newark, United States

^d Graduate School of Biomedical Sciences, Rutgers University, Newark, United States

^e Department of Computer Science, Khansar Campus, University of Isfahan, Isfahan, Iran

ARTICLE INFO

Keywords:

Arrhythmia
Convolutional Neural Network
Deep learning
Electrocardiogram
Long Short Term Memory
SHapley additive exPlanations

ABSTRACT

Objective: Electrocardiogram (ECG) is the most frequent and routine diagnostic tool used for monitoring heart electrical signals and evaluating its functionality. The human heart can suffer from a variety of diseases, including cardiac arrhythmias. Arrhythmia is an irregular heart rhythm that in severe cases can lead to stroke and can be diagnosed via ECG recordings. Since early detection of cardiac arrhythmias is of great importance, computerized and automated classification and identification of these abnormal heart signals have received much attention for the past decades.

Methods: This paper introduces a light Deep Learning (DL) approach for high accuracy detection of 8 different cardiac arrhythmias and normal rhythms. To employ DL techniques, the ECG signals were preprocessed using resampling and baseline wander removal techniques. The classification was performed using an 11-layer network employing a combination of Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM).

Results: In order to evaluate the proposed technique, ECG signals are chosen from the two physionet databases, the MIT-BIH arrhythmia database and the long-term AF database. The proposed DL framework based on the combination of CNN and LSTM showed promising results than most of the state-of-the-art methods. The proposed method reaches the mean diagnostic accuracy of 98.24%.

Conclusion: A trained model for arrhythmia classification using diverse ECG signals were successfully developed and tested.

Significance: This study presents a lightweight classification technique with high diagnostic accuracy compared to other notable methods, making it a potential candidate for implementation in Holter monitor devices for arrhythmia detection. Finally, we used SHapley Additive exPlanations (SHAP), the most popular Explainable Artificial Intelligence (XAI) method to understand how our model make predictions. The results indicate that those features (ECG samples) that have contributed the most to a prediction are consonant with clinicians' decisions. Therefore, the use of interpretable models increases the trust of clinicians in AI and thus leads to decreasing the number of misdiagnoses of cardiovascular diseases.

1. Introduction

Cardiovascular disease is amongst the three main global causes of death. As reported by the World Health Organization, nearly 18 million people die each year from heart disease [1].

An electrocardiogram (ECG) is a method of measuring and recording the electrical activity of the heart. ECG signal is recorded non-invasively by placing multiple electrodes on particular points of the body. A normal heart rhythm consists of three main parts including P waves, T waves and QRS complex [2]. The occurrence of any anomalies in the rhythm and/or heart rate indicates the existence of a disorder

* Corresponding author.

E-mail addresses: n.alamatsaz@eng.ui.ac.ir (N. Alamatsaz), l.tabatabaei@eng.ui.ac.ir (L. Tabatabaei), yazdchi@eng.ui.ac.ir (M. Yazdchi), hamidrezapayan@aut.ac.ir (H. Payan), nima.alamatsaz@njit.edu (N. Alamatsaz), f.nasimi@khc.ui.ac.ir (F. Nasimi).

<https://doi.org/10.1016/j.bspc.2023.105884>

Received 21 August 2023; Received in revised form 29 November 2023; Accepted 21 December 2023

Available online 27 December 2023

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and is called arrhythmia [3]. Heart arrhythmia is one of the chronic diseases that many people are suffering from worldwide. In some cases, arrhythmias lead to stroke or sudden cardiac death and therefore endanger human life. Early detection of arrhythmias is crucial for timely intervention and potentially saving lives [4].

Arrhythmias affect a significant number of individuals globally and can cause symptoms such as inadequate blood pumping, shortness of breath, fatigue, chest pain, and loss of consciousness [4]. For individuals at risk of heart failure, remote long-term heart monitoring using a Holter device is recommended. However, manual examination and analysis of long-term ECG recordings can be time-consuming, especially when real-time diagnosis is required. Therefore, there is a growing interest in automated computer-aided methods for arrhythmia detection to assist clinicians in diagnosing and monitoring patients [5].

Recently, Deep Learning (DL) algorithms such as Convolutional Neural Networks (CNN) as an automated computer-aided method have been frequently used for different biomedical tasks and yielded encouraging results [6]. The ability of DL networks to recognize patterns and learn important features from raw data makes them suitable for classifying ECG signals [7].

Numerous methods have been proposed for arrhythmia classification using DL models, but many of them involve complex architectures with a high number of layers, which can hinder real-time processing. Therefore, there is a need for lightweight DL models that can achieve accurate arrhythmia classification in a timely manner.

AI models especially DL models are often black boxes which makes their interpretation difficult. In order to understand what are the main features (samples) that affect the output of the model and subsequently be able to deploy these models in clinical setup (gaining clinicians trust), we need Explainable AI (XAI) techniques that unravel some of these aspects [8]. SHapley Additive exPlanation (SHAP), which is a popular XAI framework, is a concept derived from cooperative game theory, has been increasingly applied in various fields, including Machine Learning (ML) and data analysis. In the context of ECG classification, Shapley values can be a valuable tool for understanding the contributions of individual features or samples in the classification process. They measure the marginal contribution of each feature (sample) by evaluating how its inclusion or exclusion affects the predictions. By assigning a value to each feature, Shapley values provide insights into the relative importance of different features and their interactions in the ECG classification task. Understanding the Shapley values can help in several ways. Firstly, it enables feature importance ranking, allowing clinicians to prioritize the most informative features for further investigation or feature selection. Secondly, it provides a transparent and interpretable explanation of the model's decision-making process, which is particularly crucial in medical applications where interpretability is essential for trust and acceptance. Thirdly, Shapley values can help identify potential biases or inconsistencies in the model's predictions by highlighting features that have disproportionate influence [9–11].

The purpose of this study is to develop an accurate light automatic diagnostic system that assists cardiologists by providing an intelligent interpretable DL method that is time-saving, cost-efficient, trustable and reduces the number of misdiagnosed arrhythmias. The proposed model deployed CNN and recurrent neural networks (RNN) such as LSTM for the detection of various cardiac arrhythmias. ECG recordings used in this study were obtained from the MIT-BIH Arrhythmia Database and the Long-term AF database which are available from PhysioNet. As a summary, the main contributions and novelty of this paper are listed as follows:

- (1) LSTM along with CNN architectures has been deployed, because the prediction depends on the whole input sequence.
- (2) We focused on creating a lightweight model to automatically detect and classify 8 different arrhythmias as well as normal sinus rhythm. The arrhythmias that were discussed in this article

are: atrial fibrillation, atrial flutter, ventricular bigeminy, Paced rhythm, Wolf Parkinson White syndrome (WPW), supraventricular tachyarrhythmia, ventricular trigeminy and ventricular tachycardia.

- (3) Shapley values are used to show the contribution or the importance of each ECG sample on the prediction of the model.

The rest of the paper is organized as follows: Related works are presented in Section 2. The proposed system architecture is illustrated in Section 3. Experimental results are illustrated and then compared with results of other works in literature in Section 4. Finally, in the conclusions section, the current condition of the suggested technique as well as the advantages and disadvantages of the work is represented.

2. Literature review

Several researchers have carried out different AI based approaches throughout the years for classification and interpretation of various cardiac arrhythmias [12–17]. These AI techniques are ML and DL based approaches. ML techniques as automated classification methods have boosted classification accuracy in recent years. Authors in [8] described the ECG waveforms by specific features and used the features as the inputs of five different ML algorithms. This proposed work for the Right Bundle Branch Block (RBBB) and Coronary artery disease has provided better results using logistic regression and support vector machine classifier and achieved an accuracy of 95%. Authors in [18] executed a segmentation operation on the ECGs existing in the MIT-BIH database. They broke down the large records into segments containing 200 samples. Subsequently, three decision trees, random forest and logistic regression multi-class classifiers were used in order to classify 3 cardiac rhythms. The average rhythm detection time is about 1 s resulting in online and real-time performance. Eventually, random forest showed more accurate and stronger performance by achieving 88.7% accuracy and 92.5% precision. Amongst numerous applied methods, DL techniques based on CNN models have been widely implemented due to their promising performance. One way of applying CNN models on ECG signals is by virtue of transfer learning. Authors in [19], proposed a perspective for accurate classification of rare arrhythmia types which are represented in 2-D image using transfer learning method, a pre-trained CNN model called VGG16 for extracting features and a v-SVM classifier for assorting. The accuracy of the their model for normal and arrhythmia signals is 87% and 93%, respectively, resulting in an average accuracy of 90.42%. Isin et al. also used a transfer learning approach and a pre-trained CNN model called AlexNet as a feature extractor. Then a simple back-propagation was applied in order to classify the signals into three different cardiac rhythms (normal, Right Bundle Branch Block, paced). This transferred DL approach functioned efficiently by reaching an accuracy of 92% on test dataset [20]. The study in [21] used ArrhyNet to detect and classify ECG signals and to overcome the issue of imbalanced dataset the utilized Synthetic Minority Over Sampling (SMOTE) technique. The accuracy of 92.73% was achieved for their classification model. Mustaqeem et al. [22] proposed a method for differentiating between normal and diseased ECG signals obtained from the UCI database. The proposed approach for the selection of the most significant features of the ECG signals was a wrapper method, built on a random forest algorithm. Afterwards, cardiac arrhythmias were classified by means of three different SVM based techniques including one-against-all (OAA), one-against-one (OAO), and error-correcting codes (ECC). After evaluating the mentioned techniques, the OAO method proved to be best suited for classifying the ECG records into 16 categories, by achieving an accuracy of 92.07% when using a 90:10 data split ratio. Hannun et al. [23] developed a 34-layer end-to-end Deep Neural Network (DNN) for categorizing 12 rhythm classes. The DNN model was conducted on a novel vast single-lead ECG database and resulted in an encouraging AUC of 0.97. Authors in [24] used KecNet model, which contains a CNN

structure with a modified convolutional layer and a symbolic parameter extraction architecture in the feature extraction part of the model to classify arrhythmias. They achieved the accuracy of 99.31%. Authors in [25] augmented a set of time-series transformations of the original signals to the original dataset and used a 1-D CNN model to classify arrhythmias using the original and transformed database. They managed to attain an accuracy of 99% without overfitting the model. Chen et al. [26] used convolutional layers to extract primary features and incorporated LSTM and attention to improve the performance of the DNN model. Zahid et al. [27] used 1D SelfONN layers to automatically learn morphological representations from ECG data and further injected temporal features based on RR interval for timing characterization. The proposed method achieves 99.15% F1-score for normal (N) beats. Authors in [28] proposed a tiny CNN classifier for real-time monitoring of ECG at the edge with the aid of the matched filter (MF) theory. There proposed classifier achieved average sensitivity of 91.90%. In the authors previous work they managed to differentiate between shockable and non-shockable ECG signals using a 9 layer DNN model with an accuracy of 99.1% [29]. Authors in [30] considered a CNN model with 5 temporal layers, two skip connections, and one spatial layer followed by two dense layers. They interpreted their DL model by Shapley values for atrial fibrillation, Grouped supraventricular tachycardia, sinus bradycardia and sinus rhythm. Sanjana, K. et al. [31] used a transfer learning approach trained with one of the tachycardia diseases called atrial fibrillation and tested with other tachycardia diseases, such as ventricular fibrillation and sinus tachycardia. They interpreted their model and realized that the model has learned the absence of P-wave which is a common feature for ventricular fibrillation and atrial fibrillation. Authors in [32] used an autoencoder and classifier both based on CNN to classify Non-ectopic beats, Supraventricular ectopic beats, Ventricular ectopic beats, Fusion Beats and Unknown Beats. Later they managed to interpret their results using Grad-CAM.

3. Arrhythmia classification methodology

In the arrhythmia detection and classification method proposed in this study, first pre-processing is done to make the data ready to be fed into DNN algorithm and at the end, the training and testing procedure using DNN algorithm is accomplished on data. The pre-processing section is made up of following steps: noise elimination and data resampling. The research is carried out in Python programming language in JupyterLab from the Anaconda distribution of Python 3.8.11 on a system with Intel Core i7 7th Gen processor with NVIDIA GeForce GTX 1070 Ti using TensorFlow and TensorFlow-GPU 2.3.0 packages.

3.1. ECG dataset

In this study, the MIT-BIH arrhythmia database and the long-term atrial fibrillation (LTAF) database were used for training and testing purposes of the proposed framework for arrhythmia classification [33]. The MIT-BIH arrhythmia database consists of 48 half-hour ECG recordings that were obtained from 47 subjects, studied by the Beth Israel Hospital Arrhythmia Laboratory. Unlike the conventional ECG recordings that are recorded by 12 leads, this database contains heart signals which were recorded by two leads (usually MLII and V1 leads) with a 360 Hz sampling frequency [34]. The LTAF database includes 24-hours ECG signals that were obtained from 84 subjects with paroxysmal atrial fibrillation. ECG signals were recorded synchronously from two leads with 128 Hz sampling frequencies [35]. These ECG recordings were annotated with details such as rhythm type, beat type, peak locations and onset and offset of a waveform by two or more professional cardiologists. These annotations were first extracted from the signals and then used in the train and test process. Each subject's ECG recordings may contain different types of arrhythmias, so by using the rhythm type annotations, all arrhythmias were excerpted from all subjects.

3.2. Preprocessing

3.2.1. Noise filtering

ECG signals are usually corrupted by different types of either low- or high-frequency noises such as baseline wander (BW), power line interference, electromyography (EMG) noise and electrode motion artifact noise. Different types of filters can be applied to remove these noises. Respiration and subject movement are the main cause of BW which is a low-frequency artefact in ECG signal recordings of a subject. In this research, following previous works, a median filter with 200 ms and 600 ms width is used [3]. The median filter is a non-linear digital filtering technique used for noise removal of images and signals whilst preserving the useful details of the signal or the image. Each record was then normalized to an amplitude range of $[-1, +1]$.

3.2.2. Resampling

The MIT-BIH ECG recordings were digitized at 360 samples per second and the ECG recordings in the LTAF database were digitized at 128 samples per second. Therefore to make use of both databases a resampling technique is being used to downsample the signals in the MIT-BIH dataset. So after the resampling process, the frequency of all the records is 128 Hz.

As previously stated, the fusion of two heterogeneous datasets was undertaken. Subsequently, a preprocessing phase was implemented to attenuate inherent noise within the data. Following this, a normalization procedure was applied to standardize the data across the combined sets. After normalization, resampling techniques were employed to harmonize the data frequency to a uniform rate of 128 Hz.

3.3. Segmentation

Segmentation of ECG recordings is performed to homogenize the length of data fed into the model. With the sampling rate of 128 Hz and an average cardiac cycle of 0.8 s, segments with 500 samples (3.9 s) seem appropriate, since most arrhythmias appear within this length. Segments were extracted using their exclusive rhythm label. These segments were aligned to 20 samples before the p-wave peak of each beat and then were extracted in an overlapping manner. As shown in Fig. 1 the segmenting window slides one beat by one beat and produces the sections. After this step, all ECG segments from both databases are combined. As shown in Table 1 segments related to normal and atrial fibrillation classes were excessively high. One effective technique for addressing class imbalance is the strategic use of class weights. Class weights assign higher weights to the minority class, allowing the model to pay more attention to its patterns and reducing bias towards the majority class. Therefore to eliminate this side effect, evaluation metrics during both the training and testing stages were weighted by the equation shown below. In Eq. (1) TD is the total number of the data and TC is the total number of the data in each class and W_c is the weight of class c . These weights are later used during the training phase.

$$W_c = \frac{TD}{TC} \quad (1)$$

3.4. Proposed model architecture

The study introduces a high-accuracy DL technique combining CNN and LSTM architectures for end-to-end diagnosis of different types of cardiac arrhythmias from raw ECG signals. The proposed model consists of 11 layers and is trained using 85% of the data, with evaluation conducted on the remaining 15% of the segmented data.

The architecture of the proposed model as illustrated in Fig. 2 is as follows: three tandem convolutional blocks that each one contains a 1-dimensional convolutional layer for extracting high-level features and an activation function, called Rectified Linear Unit (ReLU) for achieving non-linear capabilities. The output of Relu layers is then

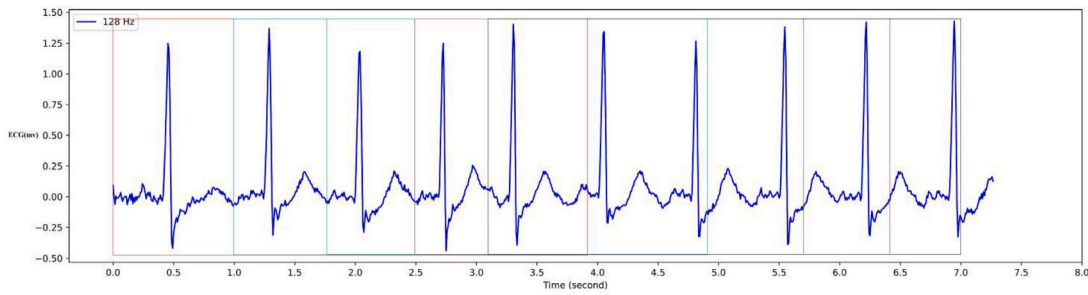


Fig. 1. Demonstrating the application of segmentation windowing technique on ECG record.

Table 1
Number of each rhythm type before and after segmentation.

Arrhythmia type	Num of rhythms in MIT-BIH	Num of rhythms in LTAf	Total num of segments
Atrial fibrillation (AFIB)	107	7358	10 000
Atrial flutter (AFL)	45	–	1099
Ventricular bigeminy (B)	221	2696	2213
Normal rhythm (N)	530	22 834	10 000
Paced rhythm(P)	60	–	7680
Wolf-Parkinson-white syndrome (PREX)	103	–	525
Supraventricular tachyarrhythmia (SVTA)	26	3268	6987
Ventricular trigeminy (T)	83	785	4752
Ventricular tachycardia (VT)	61	824	198

given to the max-pooling layer for reducing feature dimensions and selecting significant features. The output of the last convolutional block is then fed to a LSTM layer. LSTM consists of memory blocks and has a recursive feedback connection that can handle long-term dependencies and exploding gradient problems. To prevent overfitting, dropout layers are used, randomly dropping out a fraction of nodes during training. Fully connected layers usually form the last few layers of the network configuration. In this model, several dense layers were used for changing the vector's dimensions and altering the few-dimensional features to linear vectors. As the last layer multi-class activation function, softmax was used for converting the output vector of classes to probabilities for each class. Since the target variable is categorical, one-hot encoding process was used during the multi-classification procedure. Table 2 gives a detailed description of the CNN model joined by a LSTM layer implemented for this study.

4. Results and discussion

The research conducted aimed to classify eight arrhythmias and normal rhythms using a DL model. The dataset was divided into training and testing sets, with 85% of the data used for training and the remaining 15% for testing and validation. The model was compiled for 100 epochs in the training and testing set. Categorical cross-entropy was used as the loss function and the Adam optimizer was chosen because it makes the algorithm convergence faster, its implementation is straightforward, it is computationally efficient and its default hyper-parameters can be used with little tuning. Weights and biases are updated continuously using a conjugate gradient back-propagation algorithm which utilizes the generated value by the loss function until the desired optimized values are acquired. After the completion of the classification process, Shapley value analysis is done for all 9 classes of ECG arrhythmias and normal rhythm using the functions presented in [11].

4.1. Experimental results and discussion

Fig. 3 gives the confusion matrix that is used to visualize the performance of the classification algorithm. Numbers on the main diameter of this matrix indicate true positives.

Sensitivity, specificity, and Accuracy parameters are used to evaluate the proposed technique. Specificity in rhythm classification represents the proportion of true negative results among all other rhythms when a particular rhythm is considered. As shown in Fig. 4 the specificity of our model is above 90% for all 8 arrhythmias and normal rhythm, meaning that 90% of the rhythms without the specific arrhythmia are correctly classified as negative, while 10% may be false positives.

A quick review of false negative rates of AFIB, B, and T arrhythmias shown in Fig. 3 unveiled that, this poor identification of the desired arrhythmias entirely appears very sensible. In many cases, the lack of context, limited signal duration, or having a single lead limited the derivation that could be concluded from the data, making it challenging to certainly reveal whether the annotating cardiologists and/or the algorithm was correct.

Accuracy represents the proportion of correct classifications (both positive and negative) out of the total number of ECG recordings evaluated. In other words, indicates how accurately the model has performed. The average classification accuracy achieved by the proposed model was 98.24% meaning that the overall correctness or reliability of the proposed model is 98.24%. The novel 1D-CNN+LSTM model implemented in this study exhibited a high-performance accuracy for the classification of different arrhythmia types. Its implementation is straightforward and has lower computational complexity in comparison with most of the state-of-the-art approaches such as SVM classifiers-based strategies, random forest algorithm [22], or deployment of ensemble classifiers alongside SVM-based methods. Besides, the proposed model consists of a relatively low number of layers contrary to the models that were executed in [21,23–25]. Most previous research used only one database of ECG signals [4,18] but a combination of two databases with different sampling frequencies was used for training and testing procedures. Furthermore, only a limited number of arrhythmia disorders were classified in most of the earlier studies such as [18, 20,29,30] but 9 different cardiac rhythm types were distinguished in this system. In general, with increasing the number of classification classes, the model accuracy would decrease, fortunately as can be seen the proposed model can classify four more rhythms compared to the works done in [24,25] with only 0.88% accuracy loss. In the context of arrhythmia classification, sensitivity quantifies the proportion of true positive results among all rhythms that have the specific abnormality.

Table 2
Detailed description of the proposed deep learning model.

Layers	Filter size	Kernel size	Pool size	Strides	Output dimension
Conv1D	64	50	–	–	(500,64)
MaxPool1D	–	–	20	2	(250,64)
Dropout		Dropout rate = 0.1			(250,64)
Conv1D	32	10	–	–	(250,32)
MaxPool1D	–	–	10	2	(125,32)
Dropout		Dropout rate = 0.1			(125,32)
Conv1D	16	5	–	–	(125,16)
MaxPool1D	–	–	5	2	(63,16)
Dropout		Dropout rate = 0.1			(63,16)
LSTM		units=32			(32)
Dense		units=32			(32)
Dropout		Dropout rate = 0.1			(32)
Dense		units=16			(16)
softmax		output=9			(9)

Table 3
Summary of automated arrhythmia detection techniques.

Ref(work)	Database	Classification technique	Number of Layers	Accuracy	Sensitivity	Specificity	Num of classes
P.Lu et al. [24]	MIT-BIH	Light Neural Network	15	99.31%	99.45%	–	5
S.Aphale et al. [21]	MIT-BIH	CNN(ArrhyNet)	15	92.73%	–	–	5
A.Isin et al. [20]	MIT-BIH	Transferred deep CNN	8	92%	–	–	3
P.kanani et al. [25]	MIT-BIH	Time series and Deep learning	22	99.12%	–	–	5
S.Ilbeigipour et al. [18]	MIT-BIH	Structured Streaming Module	–	88.7%	83.8%	97.8%	3
A.Mustaqeem et al. [22]	Irvine Machine Learning Data Repository	Support Vector Machine(SVM)	–	92.07%	–	–	16
A.Hannun et al. [23]	Recorded by the Zio® monitor	DNN	34	85%	76.5%	97%	12
E.cimen et al. [19]	MIT-BIH	Transfer Learning	18	90.42%	–	–	2
A.Anand et al. [30]	PTB-XL	Spatio-Temporal CNN	8	89.73%	–	–	5
Proposed	MIT-BIH & LTAF	1D-CNN + LSTM	11	98.24%	86.1%	97.5%	9

As shown in Fig. 4 the sensitivity of our model is above 80% for 6 rhythms meaning that 80% of the ECGs with the specific abnormality are correctly classified as positive, while 20% may be false negatives.

The performance comparison with other recent research studies about the same problem is given in Table 3.

ROC curve is an evaluation metric that gives a graphical illustration of a classifier diagnostic capability. It is actually generated by plotting the true positive rate (TPR) also known as sensitivity against the false positive rate (FPR) also known as (1-specificity) at different threshold settings. The Area Under the Curve (AUC) is the measure that shows how well the classifier has discriminated between classes. Hence, the closer the AUC is to 1, the better the performance of the model. As it is shown in Fig. 5, the model reached nearly perfect AUC in distinguishing between each arrhythmia class.

The high computational complexity of CNNs presents a critical challenge towards their broader adoption in real-time and power-efficient scenarios. In the following, to prove that our model is lightweight and could be used in Holter monitor devices, we will show the model size and inference time to classify a single rhythm. The inference time of the proposed model on Raspberry Pi is only 5.127 ms, meaning that it takes 5.127 ms to classify one rhythm. The size of the model being loaded on the Raspberry Pi (processor being used in Holter monitors) is 0.16 MB. This implies that our model is competitive on Raspberry Pi-based Holter monitor devices.

4.2. Explainability of the proposed model

By examining the Shapley values, valuable insights can be gained regarding the relative importance of different features (samples) in the classification process and Consequently the differential diagnosis of arrhythmias. These insights can assist in distinguishing between various types of arrhythmias based on their characteristic ECG patterns. So we used the functions proposed in [11] to show the contribution or the importance of each feature (sample) on the prediction of the model. Features (samples) with larger Shapley values suggest a stronger impact on the prediction, while smaller values indicate less significance. To analyze the Shapley values the portion of the dataset that the model

has not seen is used. As shown in Table 1 the data set consists of 9 classes. In this subsection, we have shown a visual representation of the Shapley values assigned to various samples of the ECG signals. Each Figs. 6 to 14 corresponds to a specific class and in each figure red dots correspond to the top 250 Shapley values (influential samples in determining the prediction) while the Blue dots represent samples with lesser significance. According to the work presented in [30], the top 50% of the total Shapley values are generally considered more useful. Here is a summary of the findings for each class:

4.2.1. Ventricular tachycardia

ECG characteristics of ventricular tachycardia (bidirectional VT) from a diagnostic view are the absence of P wave, QRS slurring, and the opposite direction of T wave compared to QRS complex [2]. In Fig. 6 the portions of the signal highlighted by our SHAP method are the isoelectric line before the onset of the QRS complex which indicates the absence of the P wave, portions of the T wave which indicates its opposite direction, and the onset of QRS complex which indicates its slurring manner. As can be seen, the model findings are consistent with the diagnostic criteria of Ventricular Tachycardia.

4.2.2. Ventricular trigeminy

This type of abnormal heart rhythm resembles a three-beat pattern, two normal beats before a premature ventricular contraction (PVC) beat [2]. A PVC occurs when ventricles beat slightly out of rhythm. The diagnostic criteria to detect a ventricular trigeminy in an ECG signal is the detection of a PVC beat after two normal beats [36]. A PVC beat usually has a wide QRS complex with an opposite direction T wave. In Fig. 7 the SHAP method highlighted multiple portions. As can be seen, the highlighted portions show a wide QRS complex, a T wave that signifies its opposite direction and of course, it highlights the isoelectric line which shows the search for the premature beat.

4.2.3. Supraventricular tachyarrhythmia

ECG characteristics of SVTA include a narrow QRS complex with undetectable or negative P waves [2,37]. As shown in Fig. 8 our model is capable of identifying the samples related to negative P waves

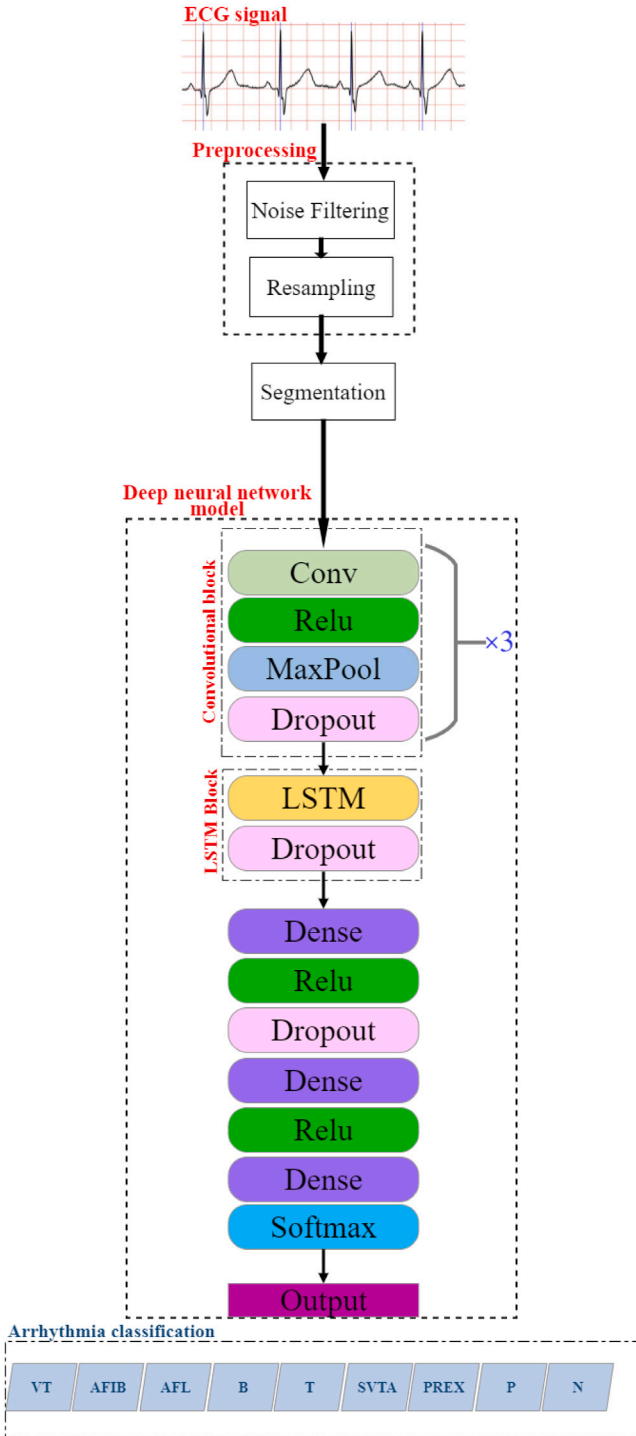


Fig. 2. Architecture of the proposed deep learning model.

or absence of P waves, also by highlighting some samples in a QRS complex can detect its narrowness. So there is a consistency between the SVTA diagnostic criteria and our findings.

4.2.4. Wolf-Parkinson-white syndrome (PREX)

Diagnostic criteria of WPW rely on two important features: small PR interval and a slurring of the initial segment of the QRS complex, known as a delta wave [2]. Fig. 9 shows that our model classifies a WPW arrhythmia correctly by focusing on portions of the ECG signal which shows the small PR interval and the delta wave, meaning that

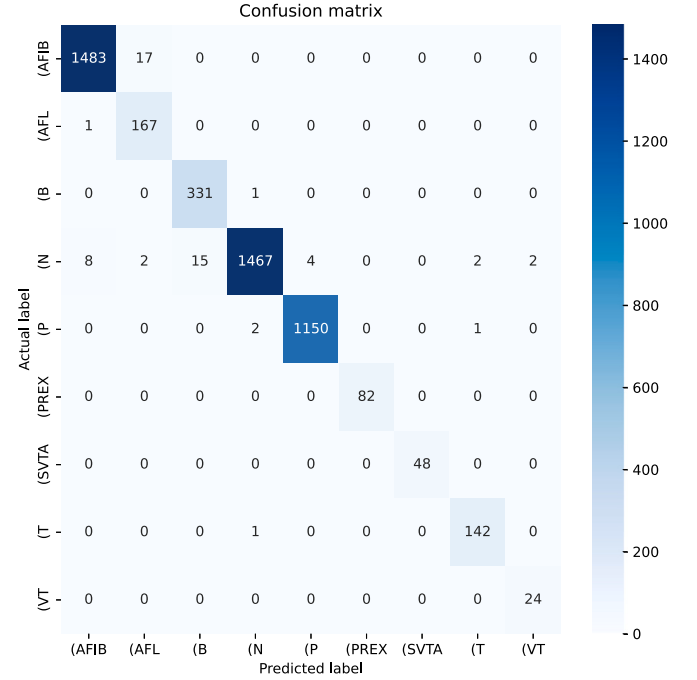


Fig. 3. Confusion matrix for the predictions of the proposed method.

our model's reason for that particular classification is consistent with the cardiologist's reasons.

4.2.5. Atrial fibrillation

Atrial fibrillation is characterized by the absence of discernible P-waves, irregular R-R intervals, and irregular baseline [2]. As shown in Fig. 10, in harmony with the opinion of the cardiologist, Shapley values interpret that the impact of highlighted portions which are rapid fibrillation waves, the isoelectric line that shows the absence of p waves, and of course the T waves which are used for detecting irregularity in ventricular response, affect the prediction more.

4.2.6. Atrial flutter

ECG features of atrial flutter are Narrow QRS complex, loss of the isoelectric baseline, and Saw-tooth pattern of inverted flutter waves [2]. As can be seen in Fig. 11 our model mostly uses the highlighted features to classify an ECG signal with AFL features as an AFL arrhythmia. The highlighted samples on the R wave represent the narrowness of the QRS complex. The highlighted portion of the signal which is shown by the (b) symbol in Fig. 11 indicates the negative flutter waves as well as the loss of the isoelectric line.

4.2.7. Ventricular bigeminy

In ventricular bigeminy, each abnormal QRS complex regularly alternates with a normal QRS complex, So a cardiologist searches for a bizarre, wide QRS complex that appears earlier than anticipated, and a T wave pointing in the opposite direction which comes after a normal sinus beat [2,36]. Samples highlighted in Fig. 12 exactly emphasize the features of a PVC which are broad QRS complex, discordant T wave, and the detection of sinus beats based on the difference in QRS complex width between PVCs and normal sinus beats.

4.2.8. Paced rhythm

Pacemaker rhythms are identified by the presence of vertical spikes of short duration [2,38]. So these spikes are highlighted by our model (Fig. 13) meaning that these samples are more important to classify a particular rhythm as a paced rhythm.

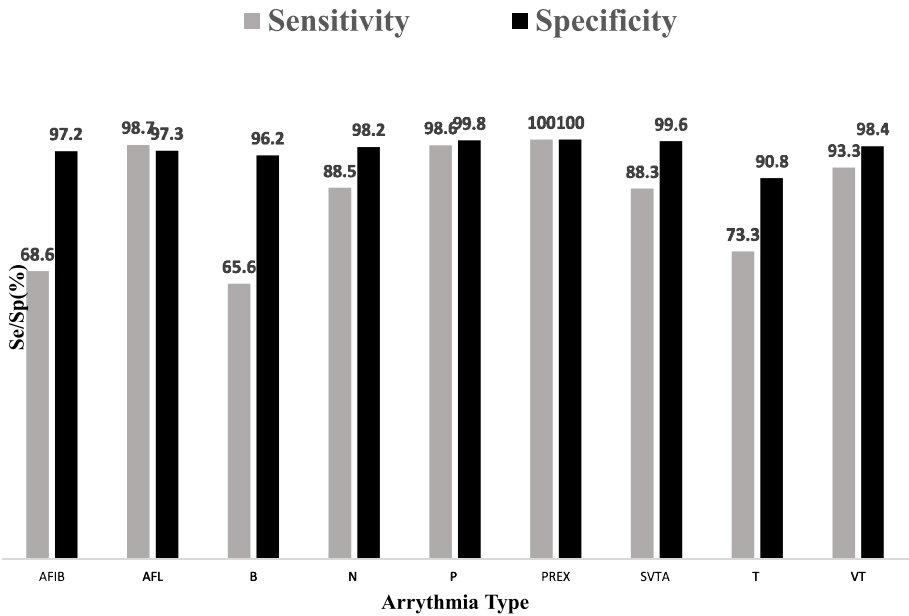


Fig. 4. Evaluation results of the proposed technique.

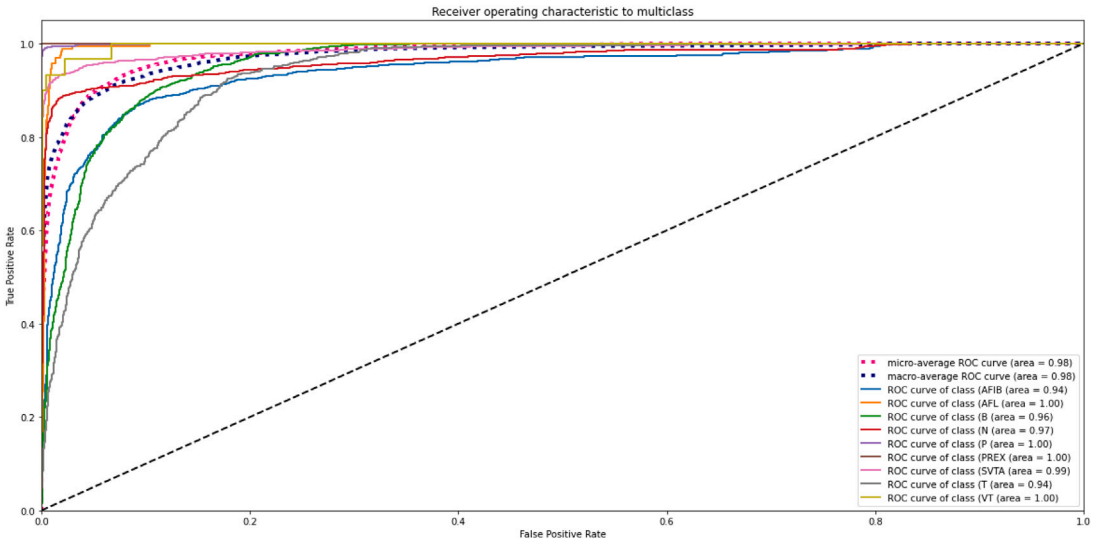


Fig. 5. Receiver operating characteristic curves for proposed model predictions on 9 rhythm classes.

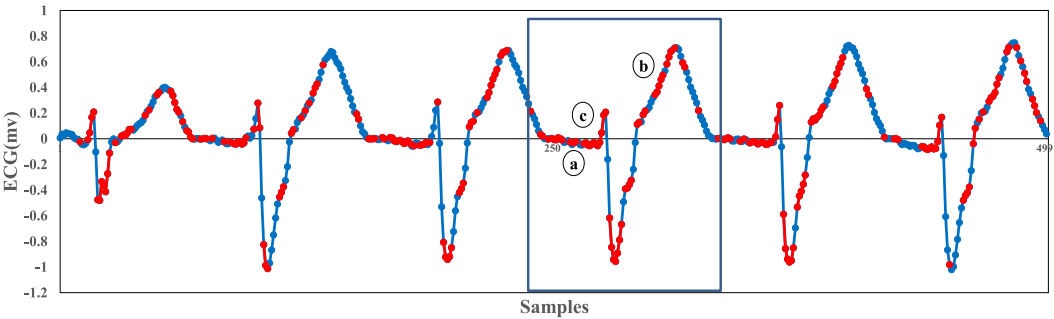


Fig. 6. Shapley Values Analysis for Ventricular Tachycardia Class: (a) Absence of P wave, (b) QRS slurring, (c) Discordant T wave.

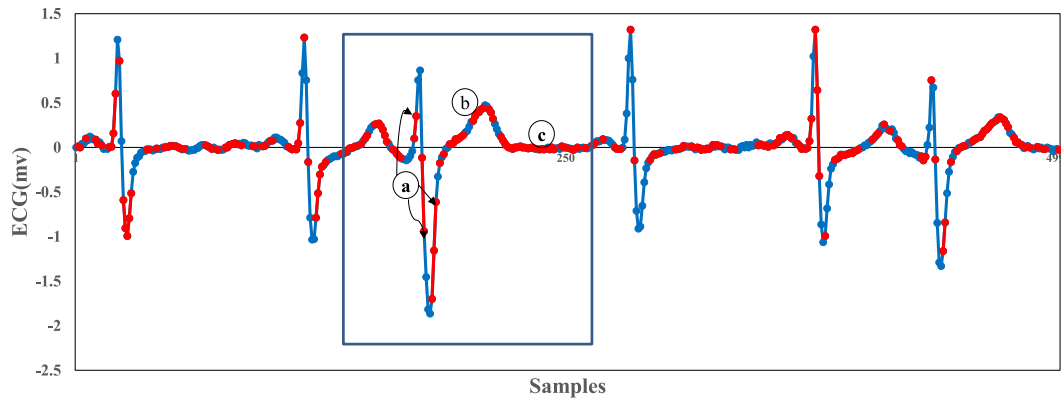


Fig. 7. Shapley Values Analysis for Ventricular Trigeminy Class: (a) Wide QRS complex, (b) Discordant T wave, (c) Isoelectric line.

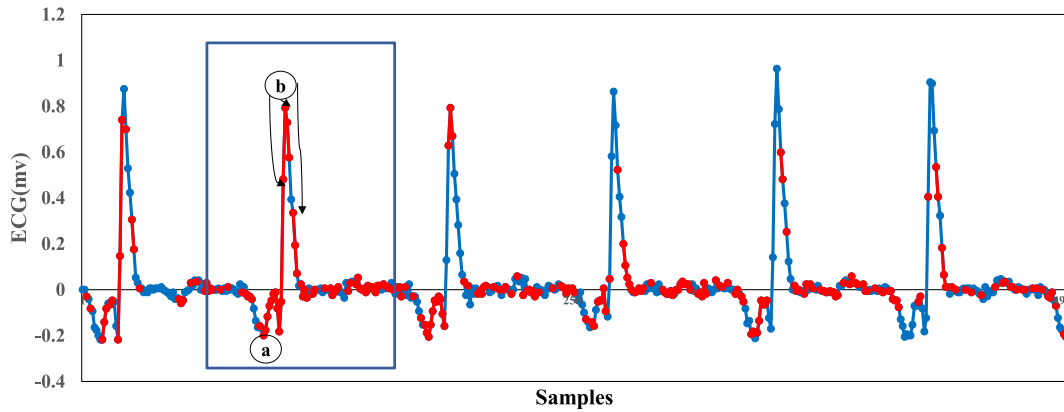


Fig. 8. Shapley Values Analysis for Supraventricular tachyarrhythmia Class: (a) Negative P waves or absence of P waves, (b) Narrow QRS complex.

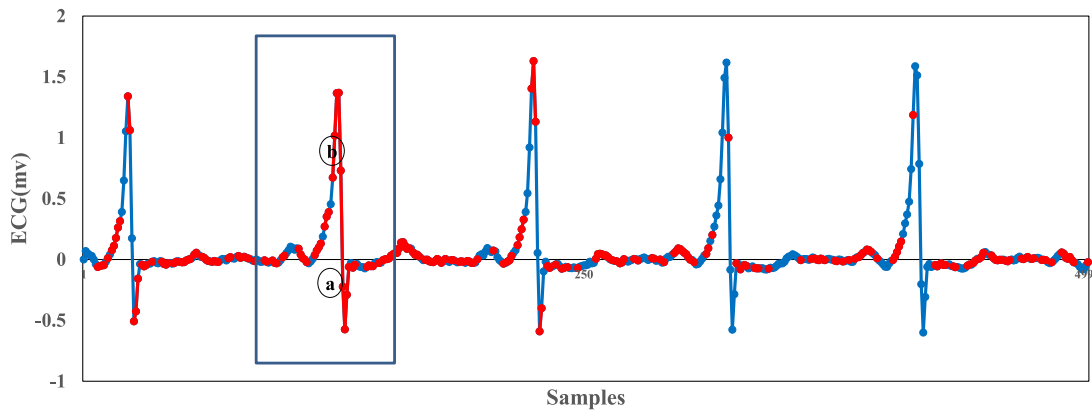


Fig. 9. Shapley Values Analysis for Wolf-Parkinson-white syndrome Class: (a) Small PR interval, (b) Delta wave.

4.2.9. Normal rhythm

ECG features of normal sinus rhythm are regular rhythm, a normal upright p wave(lead two) before each QRS complex, and constant PR interval [2]. By using Shapley values we realized that our model is also able to use the mentioned features to classify a particular rhythm correctly as a normal rhythm. As shown in Fig. 14 these portions are highlighted in red.

By examining the Shapley values of multiple relevant features, clinicians can gain a more nuanced understanding of how different arrhythmias are distinguished by the model. This information can aid in the differential diagnosis process and potentially guide further diagnostic testing or treatment decisions.

5. Conclusion and future work

Accurate detection of cardiac arrhythmias is crucial for early treatment, and computer-aided diagnosis can play a significant role in this process. In this paper, the experiment has been conducted on ECG recordings obtained from the two databases MIT-BIH and Long-term AF. The proposed CNN+LSTM model can perform the classification of 8 different types of arrhythmias and normal ECG signals. The model can extract discriminant features and information of the heart signals through CNNs and temporal features through a LSTM layer. The experiment reached an average testing accuracy of 98.24% with the inference time of 5.127 ms for a single unseen rhythm.

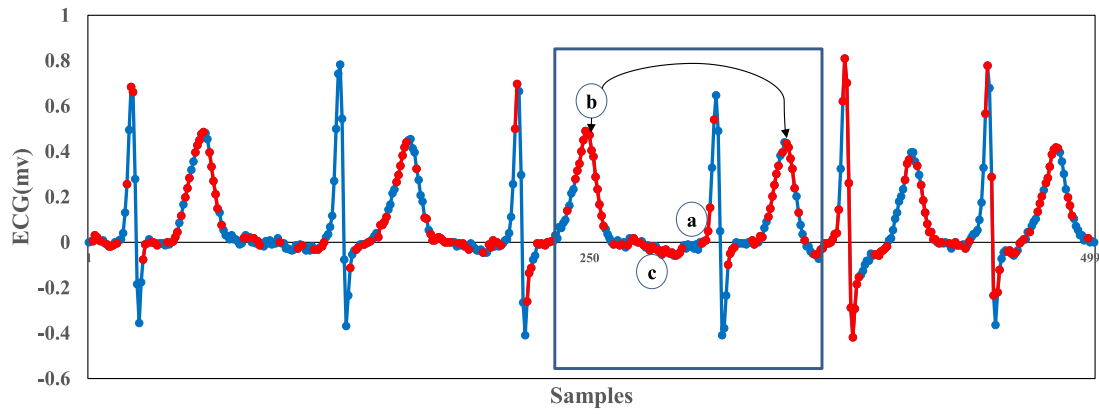


Fig. 10. Shapley Values Analysis for Atrial Fibrillation Class: (a) Absence of P-waves, (b) Irregular ventricular response, (c) Irregular baseline.

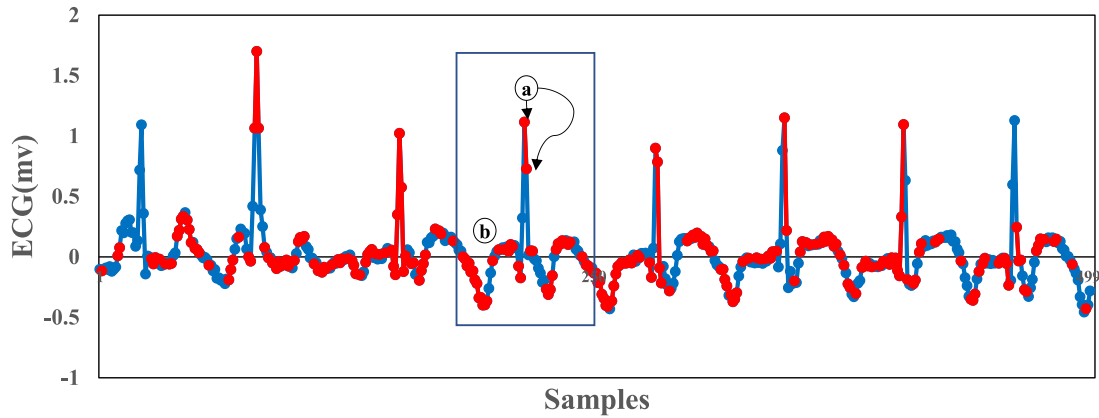


Fig. 11. Shapley Values Analysis for Atrial Flutter Class: (a) Narrow QRS complex, (b) Loss of the isoelectric baseline and Saw-tooth pattern of inverted flutter waves.

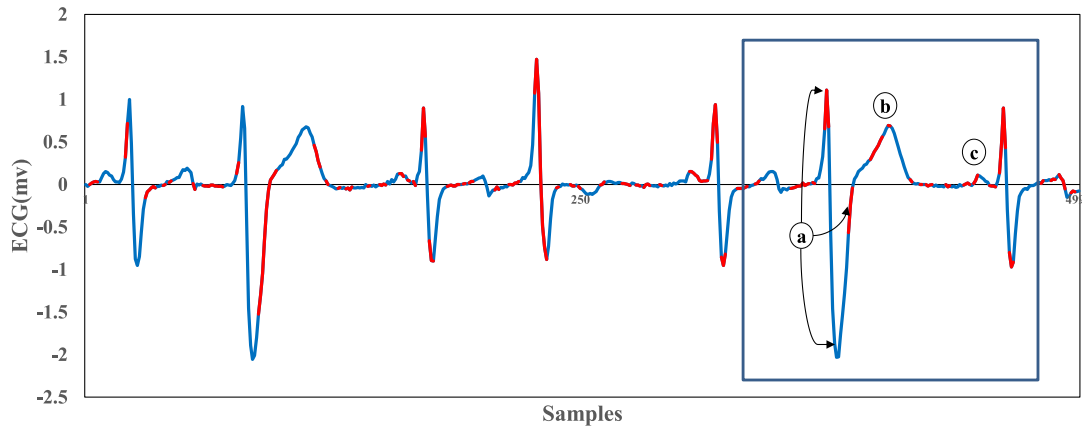


Fig. 12. Shapley Values Analysis for Ventricular bigeminy Class: (a) Wide QRS complex, (b) Discordant T wave, (c) Sinus beats.

Furthermore, the SHAP method was used to enhance the transparency and explainability of the classification process. It is highlighted that there is consonance between cardiologists' opinions and ECG features(samples) or patterns underlined by Shapley values. The use of Shapley values in ECG classification enables a more comprehensive understanding of the features driving the prediction, allowing for better-informed decision-making and potentially leading to improved diagnostic accuracy and patient care.

In addition to using the shape of the heartbeat to detect arrhythmia, utilizing the other changes in the property of the signal for training the model could be beneficial to achieving better performance. Some of these attributes are RR intervals and QRS intervals (period). For this

model, CNNs were followed by an LSTM layer to gain better accuracy in arrhythmia detection. However, the classification accuracy of some of the arrhythmias still needs improving. Deploying some rule-based algorithms alongside this model can lead to better performance. Most of the previous research has been conducted on only one database (mostly MIT-BIH) but this research was based on the combination of two databases. However, the combination of more databases is required for more reliable results. The ECG signals used in this study for training and testing the model were obtained by two leads (usually MLII and V1 leads), while in clinical applications 12-lead ECGs are known as standard. An ideal model should be able to discriminate against the standard ECG signals. Although the model was able to classify 8 types

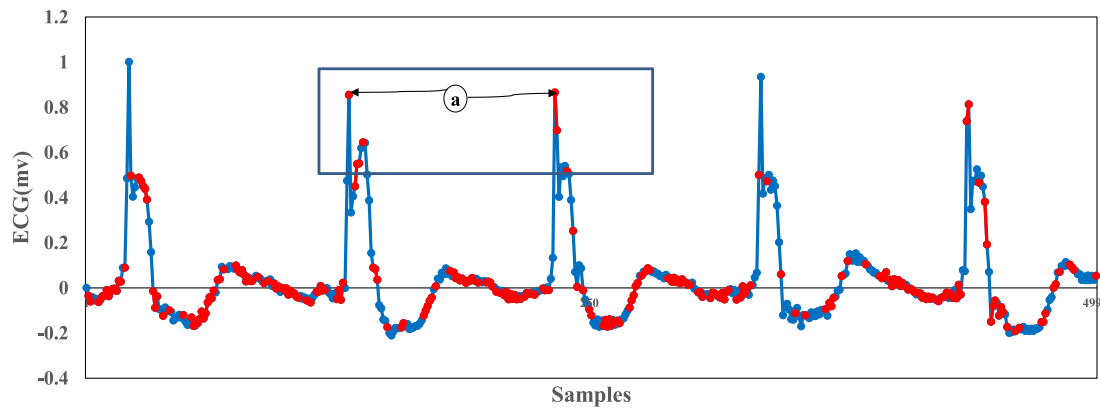


Fig. 13. Shapley Values Analysis for Paced rhythm Class: (a) Vertical spikes.

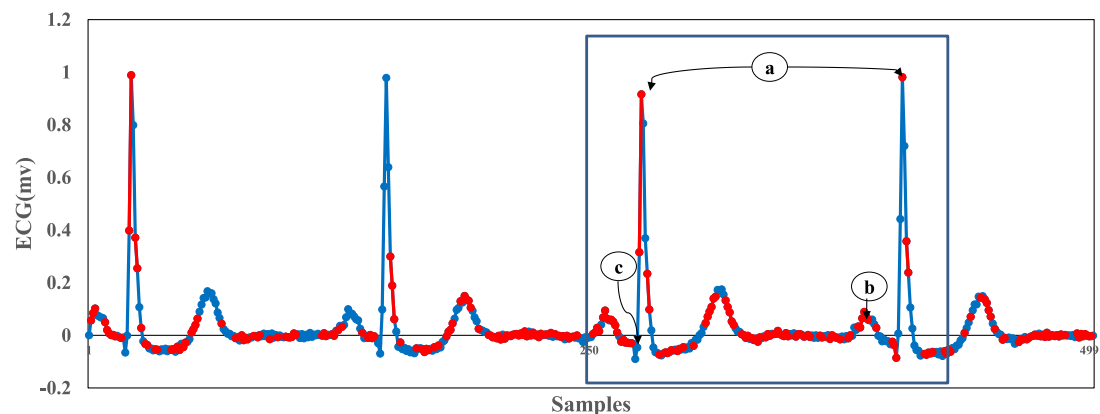


Fig. 14. Shapley Values Analysis for Normal rhythm Class: (a) Regular rhythm, (b) Normal upright p wave, (c) Constant PR interval.

of cardiac arrhythmias as well as the normal sinus rhythm with high accuracy, which is more than what some of the related works have been capable of, there still are various important undetected arrhythmias and heart disorders. The implemented model should be improved to be able to distinguish between more arrhythmias.

At the end, it is essential to note that while ML algorithms can assist in ECG classification, the final interpretation and diagnosis should always be made by a qualified healthcare professional. Human expertise and clinical judgment are still crucial in evaluating ECG findings within the context of the patient's medical history and other diagnostic tests.

CRedit authorship contribution statement

Negin Alamatsaz: Methodology, Software, Writing – original draft. **Leyla Tabatabaei:** Methodology, Software, Writing – original draft. **Mohammadreza Yazdchi:** Project administration, Resources, Supervision. **Hamidreza Payan:** Software, Validation. **Nima Alamatsaz:** Software. **Fahimeh Nasimi:** Conceptualization, Formal analysis, Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used is referenced in the paper.

Acknowledgment

We would like to express our sincere appreciation to Dr. Zahra Teimouri, a highly esteemed cardiologist, for their invaluable contributions to this manuscript. Dr. Zahra Teimouri has been a guiding force and a source of expert knowledge throughout the research process. Their deep understanding of cardiovascular medicine, extensive experience in the field, and insightful inputs have greatly enriched the quality and clinical relevance of this work.

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